Exploratory Data Analysis (EDA)

OULAD — Time-Aware Explainable Early At-Risk Prediction · Data Tasks

Group 1 – DSP391m

FPT University

Data Science Capstone (DSP391m) ▪ Report 2

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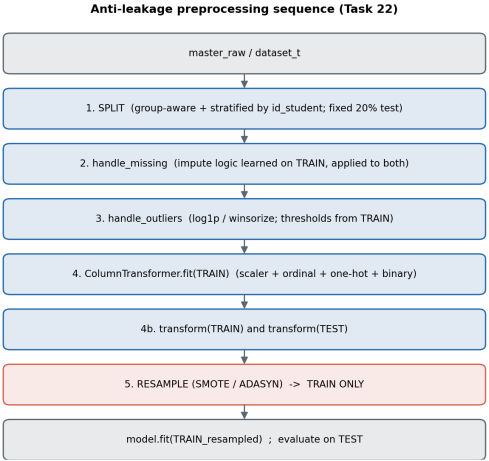
Where EDA Sits in the Pipeline

- EDA runs **after cleaning**, on the analysis-ready `master_raw` (**32,593** $\times$ **33**).
- Inputs: `master_raw` (t=100%) for univariate / bivariate / correlation; the **6 checkpoint datasets** for the time-aware analysis.
- EDA is not decoration — it **feeds three downstream decisions**:
  - feature handling (which to `log1p` / keep),
  - the **leakage check** (no $|r| \geq 0.95$),
  - the **RQ1 checkpoint schedule** (when the signal is usable).

▶ **Key:** descriptive → univariate → bivariate → multivariate → time-aware.



EDA informs steps 2–5 of the preprocessing sequence.

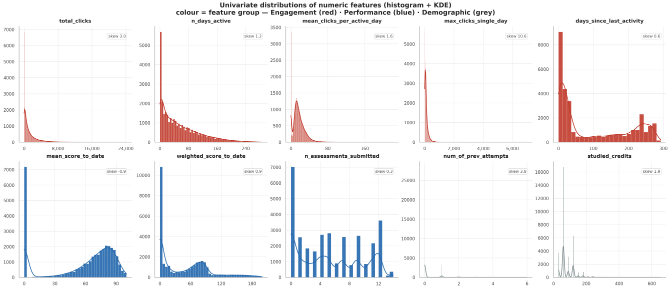
EDA Objectives & Data Profile

- **Understand** the data: 32,593 students, 33 variables, three groups — Demographic ▪ Engagement ▪ Performance.
- **Check quality**: missing values, variable types, outliers.
- **Find** features that separate at-risk vs not-at-risk students (the core question).
- **Serve two decisions**: feature selection & leakage control.

▶ **Key:** 32,593 records ▪ 33 columns ▪ 3 feature groups; quality after cleaning: **0 NaN** in feature columns, 0 duplicate keys.

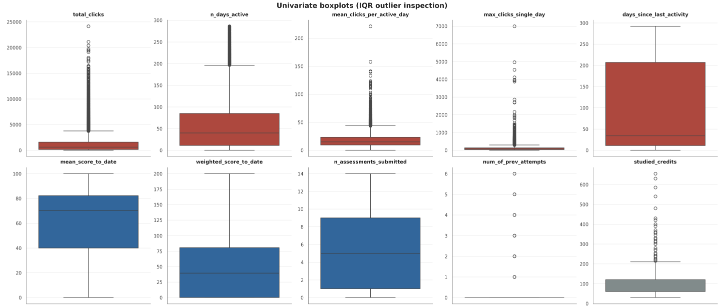
Outline

Numeric Distributions (Histogram + KDE)



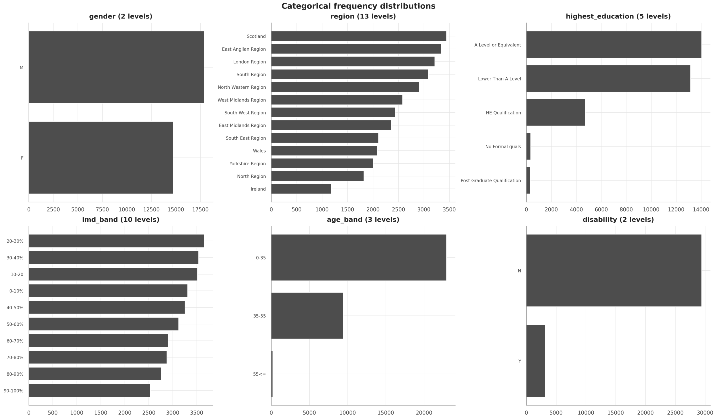
- Most clickstream features are **strongly right-skewed**: `clicks_resource` skew 34.7, kurtosis ≈2,125.
- Heavy tails are **real** (super-active students) ⇒ `log1p`, not deletion.
- `days_since_last_activity` is **bimodal** — the disengagement signature.
- Non-normal ⇒ non-parametric **Mann–Whitney** downstream.

Boxplots — Spread & Skew



- IQR rule confirms long right tails on every click feature.
- `max_clicks_single_day` reaches 6,988 vs a small median.
- Treatment: `log1p` (clicks) ▪ winsorise 1% (bounded) ▪ none (already bounded) — **no row deleted**.

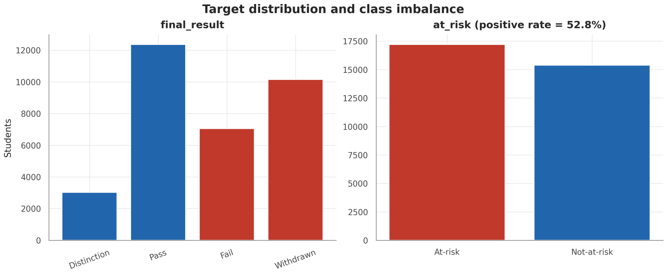
Categorical Frequency



- **83%** of students hold A-Level or below.
- Post-Graduate & No-Formal are **rare** (<1.1% each).
- Rare levels ⇒ one-hot with `handle_unknown=ignore`, not dropping.

Imbalanced categoricals shape the encoding choice (Part 4).

Target Distribution & Class Balance

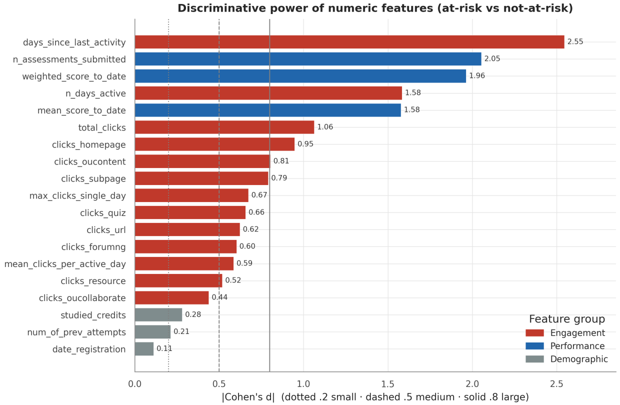


`at_risk = final_result` $\in$ `{Fail, Withdrawn}`

- At-risk **52.8%** (17,208) vs not-at-risk 47.2% (15,385) — **mild** imbalance (ratio 1.12).
- **Withdrawn** is the largest single class.
- Metrics: **PR-AUC** + **recall** on the at-risk class.

Outline

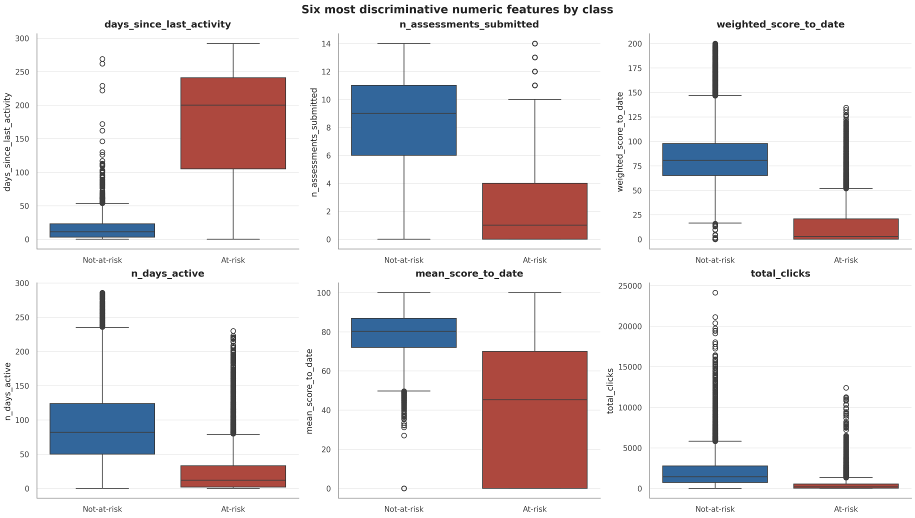
Numeric Features vs At-Risk (Cohen's *d*)



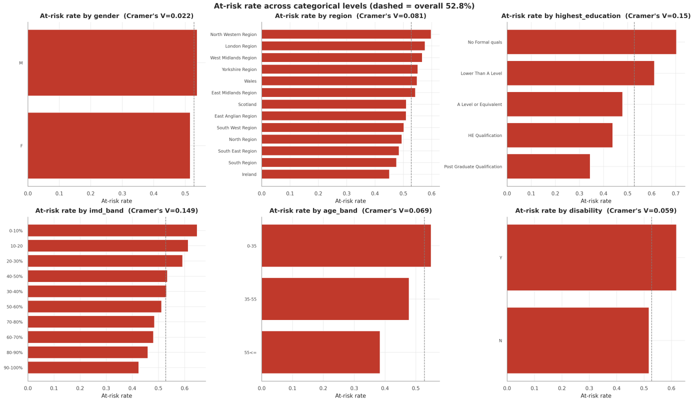
- Effect size (Cohen's *d*) + Mann–Whitney (BH-corrected).
- Strongest: `days_since_last_activity` $d = 2.55$ (171 vs 14 days), `n_assessments` 2.05 (2.3 vs 8.6), `weighted_score` 1.96.
- **All 19/19** significant ($q < 0.05$) — rank by effect size, not p.

Behaviour & performance, not demographics, drive risk.

The Strongest Features by Class



Categorical Features vs At-Risk (Cramér's V)

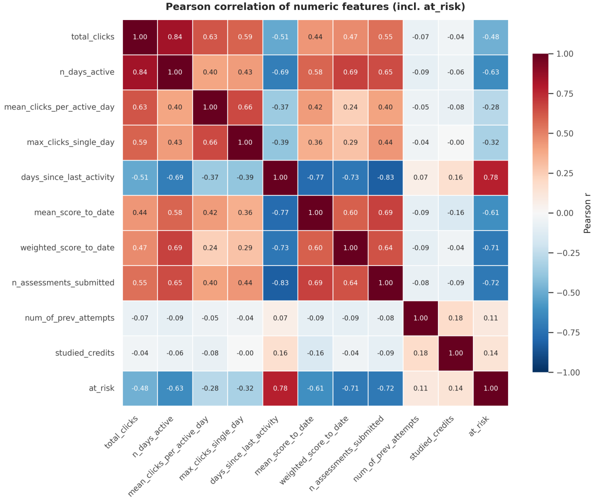


- All significant, but associations are **weak**: $V \leq 0.15$.
- Strongest: `highest_education` & `imd_band` (0.15); `gender` negligible (0.02).
- Keep demographics for **fairness** analysis, not prediction.

Demographic effect ($V \leq 0.15$) $\ll$ behavioural effect (d up to 2.55).

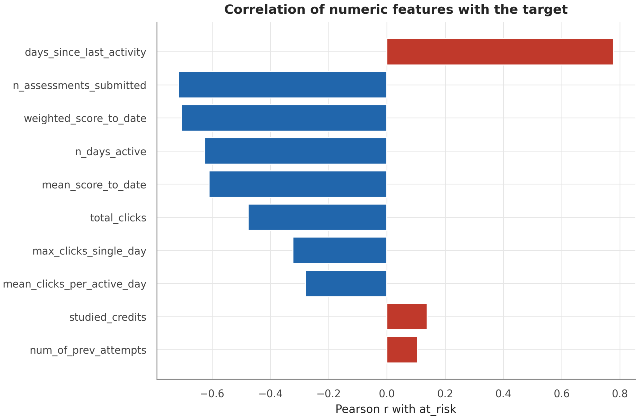
Outline

Correlation & Multicollinearity



- Pearson (linear) + Spearman (monotonic) agree on the structure.
- Two multicollinear pairs $|r| \geq 0.8$: `n_days_active–total_clicks` (0.84), `days_idle–n_assessments` (−0.83).
- Flagged for **RQ2** — SHAP/LIME may split importance among correlated features.

Correlation with the Target & Leakage Check

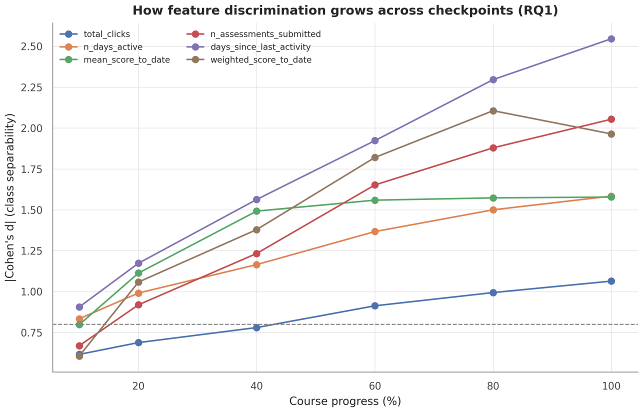


- Top $|r|$ with label: `days_idle` $+0.78$, `n_assessments` -0.72, `weighted_score` -0.71.
- Matches the Cohen's *d* ranking — two independent methods, same conclusion.
- **No feature** $|r| \geq 0.95 \Rightarrow$ no leakage suspect.

The leakage check here is the EDA half of the anti-leakage guarantee tested in the split stage.

Outline

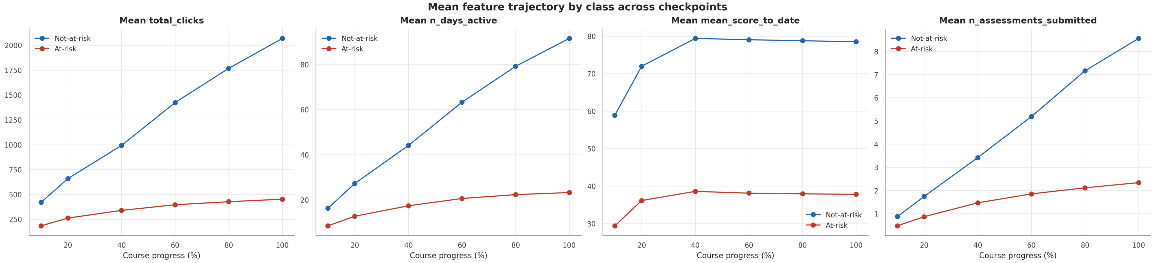
When Does the Signal Emerge?



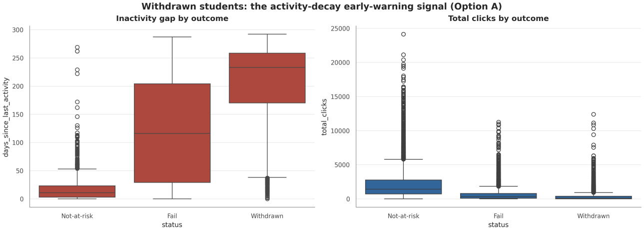
- |Cohen's *d*| per feature across the **6 checkpoints** (10%→100%); discrimination **grows steadily**.
- First to reach $d \geq 0.8$: `n_days_active` **and** `days_since_last_activity` from **10%**; scores/submissions from **20%**.

▶ **Key:** early prediction feasible from ∼**10–20%** — this sets the RQ1 checkpoint schedule.

Mean Trajectory by Class



The Withdrawn Early-Warning Signal



- Median days idle: **11** (not-at-risk) $\rightarrow$ 116 (Fail) $\rightarrow$ **233** (Withdrawn); median clicks $\sim$1,425 $\rightarrow$ 89.
- A genuine signal — the activity collapse is what makes early detection feasible.

Caveat: *withdrawn students are trivially separable late $\Rightarrow$ the honest test is the early checkpoints.*

EDA — Findings & Implications

- **Distributions:** clickstream strongly right-skewed (up to 34.7) $\Rightarrow$ `log1p`; `days_idle` bimodal.
- **Behaviour $\gg$ demographics (RQ1/RQ2):** Cohen's d up to **2.55**; all 19 numeric significant; Cramér's $V \leq 0.15$.
- **No leakage:** no $|r| \geq 0.95$; 2 multicollinear pairs flagged for SHAP stability.
- **Early signal (RQ1):** usable from $\sim$**10–20%** of course length; 40–60% is a robust window.
- **Mild imbalance (RQ3):** 52.8% at-risk $\Rightarrow$ PR-AUC/recall; resampling on train only.

EDA hands the rest of the pipeline: what to transform, which features carry signal, that there is no leakage, and when to predict.

Thank you

Key references

Kuzilek, Hlosta, Zdrahal (2017), *OULAD*, Scientific Data 4:170171. Adnan et al. (2021), *IEEE Access* 9. Tomasevic et al. (2020), *Computers & Education* 143.